Chapter 5 · ER models: turn a story into constraints

A student belongs to one department. A department can have many students. A student can have several phone numbers and can enrol in several courses. These are three different facts; the diagram must preserve each.

Vocabulary. An entity is a distinguishable object. An entity set collects entities of one type. A relationship associates entities. Cardinality constraints bound how many partners an entity can have. Participation says whether every entity must participate. Total participation means at least one; it does not mean exactly one. A weak entity needs its owner's identifier together with its partial key.

R24 Read a diagram and bound relationship sizes

Translate each edge into a sentence specifying minimum and maximum partners. Check the notation legend; arrow conventions differ. For many-to-one from A to B, each A has at most one B. If A participates totally, each A has exactly one B. If B is also total, there must be enough A entities to cover all B entities. For a plain binary many-to-many relationship, there are at most |A||B| pairs; if both sides are total and there are no additional restrictions, at least max(|A|,|B|) pairs are needed. Check feasibility and zero-sized sets separately.

Worked example W24. There are 3 students and 2 clubs. In a many-to-many membership relationship with total participation on both sides, each student needs a club and each club needs a student. At least 3 pairs are needed and 6 are possible. A minimum instance is (s1,c1),(s2,c1),(s3,c2); the maximum has every student paired with every club. Change it to many-to-one from Student to Club with total Student participation: each student has exactly one club, so there are exactly 3 pairs. If Club is also total, both clubs must be represented among those three choices. One student and two clubs would make both-side total participation impossible in the many-to-one version.

Check before moving on. Write minimum and maximum partners for each side. Check whether both totals can be satisfied before giving a numeric bound.

R25 Map entities and relationships to tables

Make a table per strong entity. Store composite attributes as their components. Usually do not store derived attributes unless specified. Give a multivalued attribute its own table with the owner's key and value. Give a weak entity a table keyed by owner key plus partial key.

For one-to-many, put the one-side key into the many-side table; total many-side participation requires non-nullness. For many-to-many, use a relationship table with both identifiers and relationship attributes. For one-to-one, choose a side for the reference and enforce uniqueness; participation influences the choice. For ternary relationships, preserve the participating tuple and stated key constraints rather than reducing automatically to binary links.

Worked example W25. For Student(ID,Name,DeptID), Department(DeptID,Name), Course(CourseID,Title), give Student.DeptID a reference to Department. If every student belongs to a department, the reference is NOT NULL. Enrolment(StudentID,CourseID,Grade) represents the many-to-many course relationship, with key (StudentID,CourseID). StudentPhone(StudentID,Phone) represents a multivalued attribute, with the pair as key when duplicate phone facts are not meaningful. A weak Dependent with partial key DependentName gets Dependent(StudentID,DependentName,Age), keyed by (StudentID,DependentName). Total

participation in its identifying relationship is necessary because a dependent cannot be identified without its owner. A composite Address can be stored as separate street/city columns; a derived Age is normally computed from a stored birth date. For a ternary Supplier-Part-Project fact, retain a table containing all three IDs. Its key follows stated constraints; three separate pair tables do not generally preserve which supplier supplied which part to which project.

Check before moving on. For each table, identify its key and references. Distinguish one composite value from independently repeated values.

R26 Minimum relations and missing information

Draw the proposed tables, keys and references. Check that every original fact can be represented without inventing partners or duplicating independent multivalued information. Merge tables only when cardinality, participation and normalization permit it. Counting entity boxes alone misses multivalued attributes, weak entities and many-to-many relationships.

Worked example W26. Two independent entities E1 and E2 have a many-to-many link. E2 also has a multivalued phone attribute. The tables are E1, E2, E2Phone and Link(E1ID,E2ID): four. Total participation on both sides requires links but does not make the many-to-many relationship mergeable. If the link becomes many-to-one from E1 to E2 and E1 is total, store E2ID in E1 and remove Link. The minimum becomes three tables. For a one-to-one relationship, a UNIQUE foreign key on an appropriate side represents the upper bound; choose the side using participation and whether unmatched entities must be kept.

Check before moving on. A merge must preserve every fact and avoid duplicate independent information. Total does not mean one-to-one.

Previous-year questions, worked

Read the original question and options in the linked official paper. These are original explanations, not replacement question statements. PDF page numbers count from the first page of the downloaded file.

DA-2026-Q61 · Minimum tables

Official question, PDF page 39. Published key: **4 to 4**.

Create E1 and E2 tables for their single-valued attributes. E2's multivalued A22 needs its own table containing E2's key A21 and A22. The many-to-many R12 needs a separate table containing both entity keys. Total participation requires each entity to appear in some R12 row but does not change the many-to-many upper bound. With no extra FDs permitting a merge, the minimum is four relations.

CS-2024-set1-Q20 · Entities versus relationships

Official question, PDF page 14. Published key: **D**.

Instructor, Course, Student and Classroom are entity types. Teaches links Instructor-Course; Registration links Student-Course; Allocation links Course-Classroom; Guides links Instructor-Student. Diagram iv has exactly those endpoints and entity/relationship roles, giving D. Diagram i swaps Student and Registration roles; ii substitutes Projects; iii connects Allocation to Registration instead of Classroom. Read each edge back into the original sentence.

CS-2024-set2-Q20 · Weak-entity participation

Official question, PDF page 14. Published key: **A**.

A weak entity requires an owner to complete its identifier. Every weak-entity instance must therefore participate in its identifying relationship: total participation, A. This does not imply that every owner has a weak entity, nor that a weak entity's partial key is globally unique. The owner's key and the partial key together identify it.

Chapter practice

Original questions. Write the answer and the intermediate reasoning. Each lettered part belongs to the same problem. The separate solutions guide contains hints, steps and a final check.

P24 · Bounds and feasibility

A has 3 entities and B has 2. Both participate totally in a binary relationship. (a) Give minimum and maximum relationship pairs for many-to-many, with examples attaining both. (b) Make the relationship many-to-one from A to B. Give the pair count. Is both-side total participation feasible if A has only 1 entity?

P25 · Schema mapping

Students each belong to one department, can enrol in many courses, have several phone numbers and can own weak Dependents identified by DependentName within a student. Courses can have many students. (a) Give the tables, primary keys and references for Student, Department, Course, Phone, Enrolment and Dependent. (b) If each student instead has at most one phone, which table can disappear?

P26 · NAT plus variation

E1 and E2 are independent strong entity types with single-attribute keys. E2 has one multivalued attribute. Their relationship is many-to-many and both sides are total; no extra dependencies allow merging. (a) Give the minimum number of 3NF tables and their purposes. (b) Make the relationship many-to-one from E1 to E2 with E1 total. Recount and state the new reference location.

Before the exam

Convert the diagram into sentences before counting. For every relation in the mapping, state what one row means. Check key constraints, total participation, multivalued attributes and whether an independent entity can remain unmatched.